驗證主機LUN路徑和多重路徑組態驗證
ONTAP

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驗證主機LUN路徑和多重路徑組態驗證

驗證主機LUN路徑和多重路徑組態驗證

在進行任何移轉之前、請確認多重路徑的設定正確、而且運作正常。LUN的所有可用路徑都應為作用中。

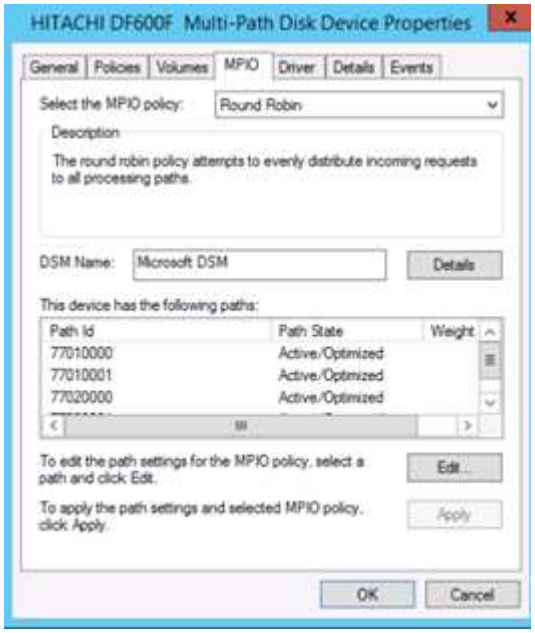
Windows主機的多重路徑驗證

在「Foreign LUN Import（FLI）（外部LUN匯入（FLI））」程序中、您應該確認主機上的多重路徑已設定且正常運作。

針對Windows主機完成下列步驟。

步驟

1. 開放式磁碟管理：
   a. 在Windows桌面上、按一下* Start*。
   b. 在「Start Search（開始搜尋）」欄位中、輸入「diskmgmt.msc」。
   c. 在「程式集」清單中、按一下「diskmgmt」。
2. 在您要驗證多個路徑的每個磁碟上按一下滑鼠右鍵、然後按一下「內容」。
3. 在MPIO索引標籤的Select the MPIO policy（選取MPIO原則）清單中、按一下所有作用中的路徑。



若要使用命令列驗證多重路徑、請完成下列步驟：

4. 開啟Windows命令提示字元。
5. 執行mpclaim.exe -v c：\multipathconfig.txt以擷取多重路徑組態。

Linux主機的多重路徑驗證

在「Foreign LUN Import（FLI）（外部LUN匯入（FLI））」程序中、您應該確認主機上的多重路徑已設定且正常運作。

完成下列Linux主機步驟。

步驟

1. 若要驗證DM-MP多重路徑在Linux主機上是否已設定且正常運作、請執行下列命令：「multiPath-ll」

```
mpath2 (360060e801046b96004f2bf4600000012) dm-6 HITACHI,DF600F
[size=2.0G][features=0][hwhandler=0][rw]
\_ round-robin 0 [prio=1][active]
 \_ 0:0:1:2 sdg 8:96  [active][ready]
 \_ 1:0:1:2 sdo 8:224 [active][ready]
\_ round-robin 0 [prio=0][enabled]
 \_ 0:0:0:2 sdc 8:32  [active][ready]
 \_ 1:0:0:2 sdk 8:160 [active][ready]
mpath1 (360060e801046b96004f2bf4600000011) dm-5 HITACHI,DF600F
[size=2.0G][features=0][hwhandler=0][rw]
\_ round-robin 0 [prio=1][active]
 \_ 0:0:0:1 sdb 8:16  [active][ready]
 \_ 1:0:0:1 sdj 8:144 [active][ready]
\_ round-robin 0 [prio=0][enabled]
 \_ 0:0:1:1 sdf 8:80  [active][ready]
 \_ 1:0:1:1 sdn 8:208 [active][ready]
mpath0 (360060e801046b96004f2bf4600000010) dm-0 HITACHI,DF600F
[size=20G][features=0][hwhandler=0][rw]
\_ round-robin 0 [prio=1][active]
 \_ 0:0:1:0 sde 8:64  [active][ready]
 \_ 1:0:1:0 sdm 8:192 [active][ready]
\_ round-robin 0 [prio=0][enabled]
 \_ 0:0:0:0 sda 8:0   [active][ready]
 \_ 1:0:0:0 sdi 8:128 [active][ready]
mpath3 (360060e801046b96004f2bf4600000013) dm-7 HITACHI,DF600F
[size=3.0G][features=0][hwhandler=0][rw]
\_ round-robin 0 [prio=1][active]
 \_ 0:0:0:3 sdd 8:48  [active][ready]
 \_ 1:0:0:3 sdl 8:176 [active][ready]
\_ round-robin 0 [prio=0][enabled]
 \_ 0:0:1:3 sdh 8:112 [active][ready]
 \_ 1:0:1:3 sdp 8:240 [active][ready]
[root@dm-rx200s6-22 ~]#
```

ESXi主機的多重路徑驗證

在「Foreign LUN Import（FLI）（外部LUN匯入（FLI））」程序中、您應該確認主機上的多重路徑已設定且正常運作。

針對ESXi主機完成下列步驟。

步驟

1. 使用VMware vSphere Client判斷ESXi和虛擬機器。

2. 使用vSphere Client判斷要移轉的SAN LUN。

3. 確定要移轉的VMFS和RDM（vfat）磁碟區：「esxcli儲存檔案系統清單」

```
Mount Point                                          Volume Name
UUID                                  Mounted  Type           Size
Free
----------------------------------------------------  -----------------
-----------------------------------  -------  ------  -----------
-----------
/vmfs/volumes/538400f6-3486df59-52e5-00262d04d700  BootLun_datastore
538400f6-3486df59-52e5-00262d04d700     true  VMFS-5  13421772800
12486443008
/vmfs/volumes/53843dea-5449e4f7-88e0-00262d04d700  VM_datastore
53843dea-5449e4f7-88e0-00262d04d700     true  VMFS-5  42681237504
6208618496
/vmfs/volumes/538400f6-781de9f7-c321-00262d04d700
538400f6-781de9f7-c321-00262d04d700     true  vfat     4293591040
4269670400
/vmfs/volumes/c49aad7f-afbab687-b54e-065116d72e55
c49aad7f-afbab687-b54e-065116d72e55     true  vfat      261853184
77844480
/vmfs/volumes/270b9371-8fbedc2b-1f3b-47293e2ce0da
270b9371-8fbedc2b-1f3b-47293e2ce0da     true  vfat      261853184
261844992
/vmfs/volumes/538400ef-647023fa-edef-00262d04d700
538400ef-647023fa-edef-00262d04d700     true  vfat      299712512
99147776
~ #
```

> 如果VMFS具有extends（跨距VMFS）、則應移轉屬於跨距的所有LUN。若要顯示GUI中的所有擴充功能、請移至「組態」→「硬體」→「儲存設備」、然後按一下「資料存放區」以選取「內容」連結。

> 移轉後、將其新增回儲存設備時、您會看到多個具有相同VMFS標籤的LUN項目。在此案例中、您應要求客戶僅選取標示為標頭的項目。

4. 確定要移轉的LUN和大小：「esxcfg-scsidevs -c」

```
Device UID                            Device Type      Console Device
Size      Multipath PluginDisplay Name
mpx.vmhba36:C0:T0:L0                  CD-ROM
/vmfs/devices/cdrom/mpx.vmhba36:C0:T0:L0                  0MB       NMP
Local Optiarc CD-ROM (mpx.vmhba36:C0:T0:L0)
naa.60060e801046b96004f2bf4600000014  Direct-Access
/vmfs/devices/disks/naa.60060e801046b96004f2bf4600000014  20480MB   NMP
HITACHI Fibre Channel Disk (naa.60060e801046b96004f2bf4600000014)
naa.60060e801046b96004f2bf4600000015  Direct-Access
/vmfs/devices/disks/naa.60060e801046b96004f2bf4600000015  40960MB   NMP
HITACHI Fibre Channel Disk (naa.60060e801046b96004f2bf4600000015)
~~~~~~ Output truncated ~~~~~~~
~ #
```

5. 識別要移轉的原始裝置對應（RDM）LUN。
6. 尋找RDM裝置：「尋找/vmfs/volumes -name **-rdm**」

```
/vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Windows2003/Windows2003_1-rdmp.vmdk
/vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Windows2003/Windows2003_2-rdm.vmdk
/vmfs/volumes/53843dea-5449e4f7-88e0-00262d04d700/Linux/Linux_1-rdm.vmdk
/vmfs/volumes/53843dea-5449e4f7-88e0-00262d04d700/Solaris10/Solaris10_1-
rdmp.vmdk
```

7. 從先前的輸出中移除-rdmp和-RDM、然後執行vmkfsools命令來尋找VML對應和RDM類型。

```
# vmkfstools -q /vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Windows2003/Windows2003_1.vmdk
vmkfstools -q /vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Windows2003/Windows2003_1.vmdk
Disk /vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Windows2003/Windows2003_1.vmdk is a Passthrough Raw Device
Mapping
Maps to: vml.020002000060060e801046b96004f2bf4600000016444636303046
~ # vmkfstools -q /vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Windows2003/Windows2003_2.vmdk
Disk /vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Windows2003/Windows2003_2.vmdk is a Non-passthrough Raw
Device Mapping
Maps to: vml.020003000060060e801046b96004f2bf4600000017444636303046
~ # vmkfstools -q /vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Linux/Linux_1.vmdk
Disk /vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Linux/Linux_1.vmdk is a Non-passthrough Raw Device Mapping
Maps to: vml.020005000060060e801046b96004f2bf4600000019444636303046
~ # vmkfstools -q /vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Solaris10/Solaris10_1.vmdk
Disk /vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Solaris10/Solaris10_1.vmdk is a Passthrough Raw Device
Mapping
Maps to: vml.020004000060060e801046b96004f2bf4600000018444636303046
~ #
```

PassthThrough為RDM、實體為（RDMP）、而非PassthThrough則為RDM、虛擬為（RDMV\）。虛擬RDM和VM Snapshot複本的VM會在移轉後中斷、因為VM Snapshot delta vmDK指向具有過時naa ID的RDM。因此在移轉之前、請客戶移除此類VM中的所有Snapshot複本。在VM上按一下滑鼠右鍵、然後按一下Snapshot（快照）→ Snapshot Manager Delete All（全部刪除）按鈕。請參閱NetApp知識庫3013935、以瞭解有關NetApp儲存設備上VMware硬體加速鎖定的詳細資訊。

8. 識別LUN naa與RDM裝置對應。

```
~ # esxcfg-scsidevs -u | grep
vml.020002000060060e801046b96004f2bf4600000016444636303046
naa.60060e801046b96004f2bf4600000016
vml.020002000060060e801046b96004f2bf4600000016444636303046
~ # esxcfg-scsidevs -u | grep
vml.020003000060060e801046b96004f2bf4600000017444636303046
naa.60060e801046b96004f2bf4600000017
vml.020003000060060e801046b96004f2bf4600000017444636303046
~ # esxcfg-scsidevs -u | grep
vml.020005000060060e801046b96004f2bf4600000019444636303046
naa.60060e801046b96004f2bf4600000019
vml.020005000060060e801046b96004f2bf4600000019444636303046
~ # esxcfg-scsidevs -u | grep
vml.020004000060060e801046b96004f2bf4600000018444636303046
naa.60060e801046b96004f2bf4600000018
vml.020004000060060e801046b96004f2bf4600000018444636303046
~ #
```

9. 確定虛擬機器組態：「esxcli儲存檔案系統清單| grep VMFS」

```
/vmfs/volumes/538400f6-3486df59-52e5-00262d04d700  BootLun_datastore
538400f6-3486df59-52e5-00262d04d700     true  VMFS-5  13421772800
12486443008
/vmfs/volumes/53843dea-5449e4f7-88e0-00262d04d700  VM_datastore
53843dea-5449e4f7-88e0-00262d04d700     true  VMFS-5  42681237504
6208618496
~ #
```

10. 記錄資料存放區的UUID。
11. 複製「/etc/vmware/hostd/vmInventory．xml」、並記下檔案和vmx組態路徑的內容。

```
~ # cp /etc/vmware/hostd/vmInventory.xml
/etc/vmware/hostd/vmInventory.xml.bef_mig
~ # cat /etc/vmware/hostd/vmInventory.xml
<ConfigRoot>
  <ConfigEntry id="0001">
    <objID>2</objID>
    <vmxCfgPath>/vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Windows2003/Windows2003.vmx</vmxCfgPath>
  </ConfigEntry>
  <ConfigEntry id="0004">
    <objID>5</objID>
    <vmxCfgPath>/vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Linux/Linux.vmx</vmxCfgPath>
  </ConfigEntry>
  <ConfigEntry id="0005">
    <objID>6</objID>
    <vmxCfgPath>/vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Solaris10/Solaris10.vmx</vmxCfgPath>
  </ConfigEntry>
</ConfigRoot>
```

12. 識別虛擬機器硬碟。

    移轉後必須提供此資訊、才能依序新增移除的RDM裝置。

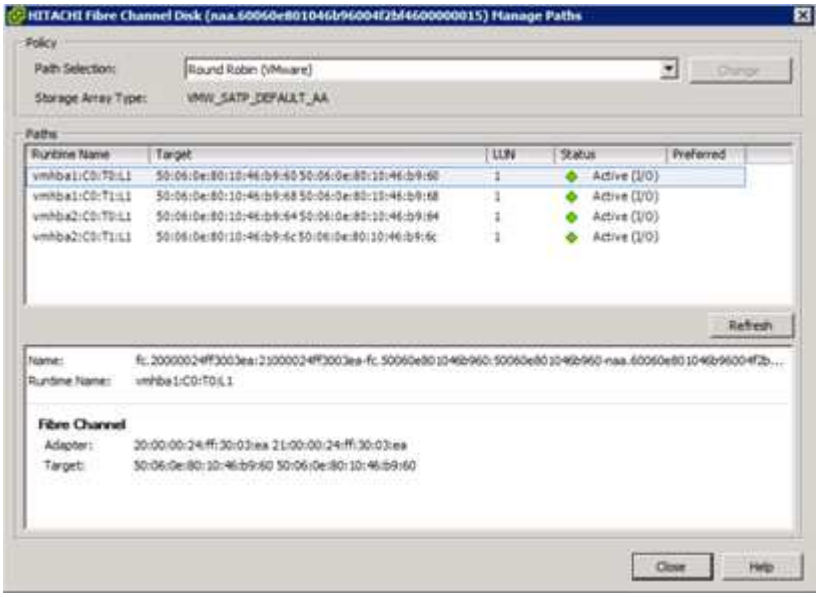
```
~ # grep fileName /vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Windows2003/Windows2003.vmx
scsi0:0.fileName = "Windows2003.vmdk"
scsi0:1.fileName = "Windows2003_1.vmdk"
scsi0:2.fileName = "Windows2003_2.vmdk"
~ # grep fileName /vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Linux/Linux.vmx
scsi0:0.fileName = "Linux.vmdk"
scsi0:1.fileName = "Linux_1.vmdk"
~ # grep fileName /vmfs/volumes/53843dea-5449e4f7-88e0-
00262d04d700/Solaris10/Solaris10.vmx
scsi0:0.fileName = "Solaris10.vmdk"
scsi0:1.fileName = "Solaris10_1.vmdk"
~ #
```

13. 判斷RDM裝置、虛擬機器對應及相容模式。
14. 使用上述資訊、記下RDM對應至裝置、虛擬機器、相容模式和順序。

    稍後將RDM裝置新增至VM時、您將需要此資訊。

```
Virtual Machine -> Hardware -> NAA -> Compatibility mode
Windows2003 VM -> scsi0:1.fileName = "Windows2003_1.vmdk" ->
naa.60060e801046b96004f2bf4600000016
-> RDM Physical
Windows2003 VM -> scsi0:2.fileName = "Windows2003_2.vmdk" ->
naa.60060e801046b96004f2bf4600000017
-> RDM Virtual
Linux VM -> scsi0:1.fileName = “Linux_1.vmdk” ->
naa.60060e801046b96004f2bf4600000019 -> RDM Virtual
Solaris10 VM -> scsi0:1.fileName = “Solaris10_1.vmdk” ->
naa.60060e801046b96004f2bf4600000018 -> RDM Physical
```

15. 確定多重路徑組態。
16. 在vSphere Client中取得儲存設備的多重路徑設定：
    a. 在vSphere Client中選取ESX或ESXi主機、然後按一下「組態」索引標籤。
    b. 按一下「儲存設備」。
    c. 選取資料存放區或對應的LUN。
    d. 按一下*「內容」*。
    e. 在「內容」對話方塊中、視需要選取所需的範圍。
    f. 按一下*「Extent Device」（範圍裝置）>*「Manage Paths」（管理路徑）*、然後在「Manage Path」（管理路徑）對話方塊中取得路徑



17. 從ESXi主機命令列取得LUN多重路徑資訊：
    a. 登入ESXi主機主控台。
    b. 執行esxcli儲存設備NMP裝置清單以取得多重路徑資訊。

```
# esxcli storage nmp device list
naa.60060e801046b96004f2bf4600000014
   Device Display Name: HITACHI Fibre Channel Disk
(naa.60060e801046b96004f2bf4600000014)
   Storage Array Type: VMW_SATP_DEFAULT_AA
   Storage Array Type Device Config: SATP VMW_SATP_DEFAULT_AA does
not support device configuration.
   Path Selection Policy: VMW_PSP_RR
   Path Selection Policy Device Config:
{policy=rr,iops=1000,bytes=10485760,useANO=0; lastPathIndex=3:
NumIOsPending=0,numBytesPending=0}
   Path Selection Policy Device Custom Config:
   Working Paths: vmhba2:C0:T1:L0, vmhba2:C0:T0:L0, vmhba1:C0:T1:L0,
vmhba1:C0:T0:L0
   Is Local SAS Device: false
   Is Boot USB Device: false

naa.60060e801046b96004f2bf4600000015
   Device Display Name: HITACHI Fibre Channel Disk
(naa.60060e801046b96004f2bf4600000015)
   Storage Array Type: VMW_SATP_DEFAULT_AA
   Storage Array Type Device Config: SATP VMW_SATP_DEFAULT_AA does
not support device configuration.
   Path Selection Policy: VMW_PSP_RR
   Path Selection Policy Device Config:
{policy=rr,iops=1000,bytes=10485760,useANO=0; lastPathIndex=0:
NumIOsPending=0,numBytesPending=0}
   Path Selection Policy Device Custom Config:
   Working Paths: vmhba2:C0:T1:L1, vmhba2:C0:T0:L1, vmhba1:C0:T1:L1,
vmhba1:C0:T0:L1
   Is Local SAS Device: false
   Is Boot USB Device: false

naa.60060e801046b96004f2bf4600000016
   Device Display Name: HITACHI Fibre Channel Disk
(naa.60060e801046b96004f2bf4600000016)
   Storage Array Type: VMW_SATP_DEFAULT_AA
   Storage Array Type Device Config: SATP VMW_SATP_DEFAULT_AA does
not support device configuration.
   Path Selection Policy: VMW_PSP_RR
   Path Selection Policy Device Config:
{policy=rr,iops=1000,bytes=10485760,useANO=0; lastPathIndex=1:
NumIOsPending=0,numBytesPending=0}
   Path Selection Policy Device Custom Config:
   Working Paths: vmhba2:C0:T1:L2, vmhba2:C0:T0:L2, vmhba1:C0:T1:L2,
vmhba1:C0:T0:L2
```

```
   Is Local SAS Device: false
   Is Boot USB Device: false

naa.60060e801046b96004f2bf4600000017
   Device Display Name: HITACHI Fibre Channel Disk
(naa.60060e801046b96004f2bf4600000017)
   Storage Array Type: VMW_SATP_DEFAULT_AA
   Storage Array Type Device Config: SATP VMW_SATP_DEFAULT_AA does
not support device configuration.
   Path Selection Policy: VMW_PSP_RR
   Path Selection Policy Device Config:
{policy=rr,iops=1000,bytes=10485760,useANO=0; lastPathIndex=1:
NumIOsPending=0,numBytesPending=0}
   Path Selection Policy Device Custom Config:
   Working Paths: vmhba2:C0:T1:L3, vmhba2:C0:T0:L3, vmhba1:C0:T1:L3,
vmhba1:C0:T0:L3
   Is Local SAS Device: false
   Is Boot USB Device: false

naa.60060e801046b96004f2bf4600000018
   Device Display Name: HITACHI Fibre Channel Disk
(naa.60060e801046b96004f2bf4600000018)
   Storage Array Type: VMW_SATP_DEFAULT_AA
   Storage Array Type Device Config: SATP VMW_SATP_DEFAULT_AA does
not support device configuration.
   Path Selection Policy: VMW_PSP_RR
   Path Selection Policy Device Config:
{policy=rr,iops=1000,bytes=10485760,useANO=0; lastPathIndex=1:
NumIOsPending=0,numBytesPending=0}
   Path Selection Policy Device Custom Config:
   Working Paths: vmhba2:C0:T1:L4, vmhba2:C0:T0:L4, vmhba1:C0:T1:L4,
vmhba1:C0:T0:L4
   Is Local SAS Device: false
   Is Boot USB Device: false

naa.60060e801046b96004f2bf4600000019
   Device Display Name: HITACHI Fibre Channel Disk
(naa.60060e801046b96004f2bf4600000019)
   Storage Array Type: VMW_SATP_DEFAULT_AA
   Storage Array Type Device Config: SATP VMW_SATP_DEFAULT_AA does
not support device configuration.
   Path Selection Policy: VMW_PSP_RR
   Path Selection Policy Device Config:
{policy=rr,iops=1000,bytes=10485760,useANO=0; lastPathIndex=1:
NumIOsPending=0,numBytesPending=0}
   Path Selection Policy Device Custom Config:
```

```
   Working Paths: vmhba2:C0:T1:L5, vmhba2:C0:T0:L5, vmhba1:C0:T1:L5,
vmhba1:C0:T0:L5
   Is Local SAS Device: false
   Is Boot USB Device: false
```

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